Props: Verifiable Machine-Learning Inference over Private Data

Dani Vilardell*
Cornell Tech, IC3

Samuel Breckenridge*
Cornell Tech, IC3

Derek Leung
MIT

Andrés Fábrega
Cornell Tech, IC3

James Austgen
Cornell Tech, IC3

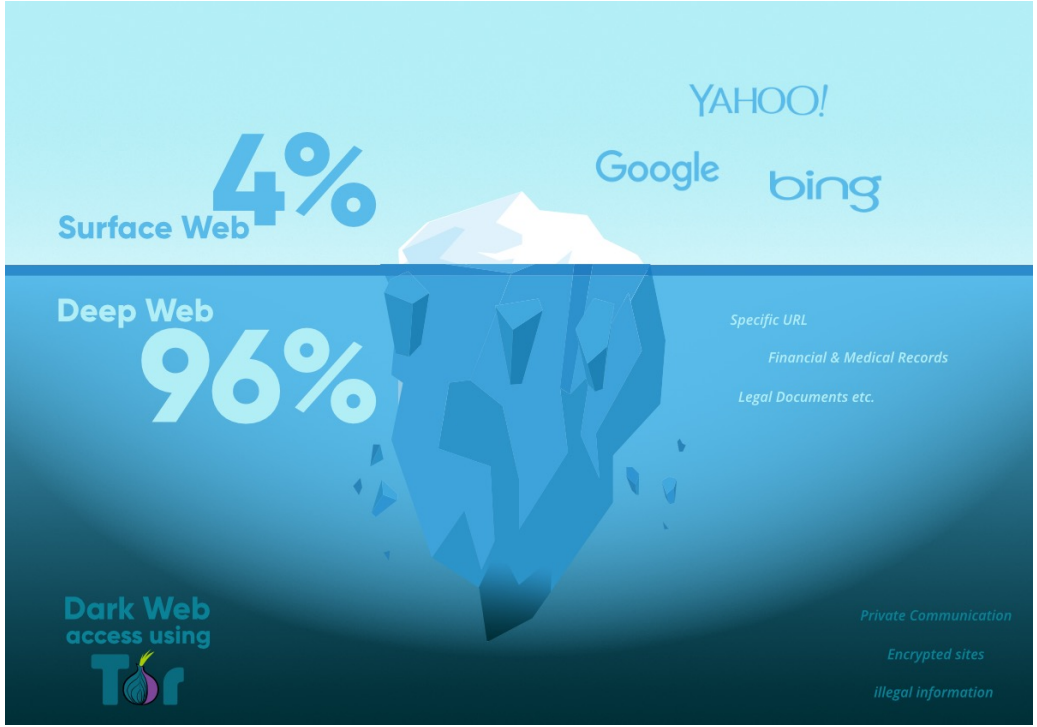
Jung-Woo Chang
UCSD

Sergey Gorbunov
Interop Labs

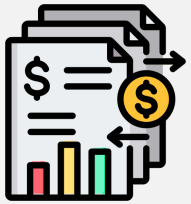
Farinaz Koushanfar
UCSD

Ari Juels
Cornell Tech, IC3

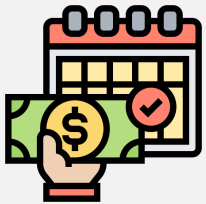
We live in a private web world



Deep web data contains information about

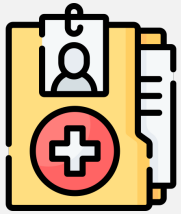


BEHAVIORAL PATTERNS
(e.g. Transaction histories)



FINANCIAL ATTRIBUTES
(e.g. Income stability)

VERIFIED OWNERSHIP
(e.g. Warranty deed)



PERSONAL RECORDS
(e.g. medical records or emails)

The "Private Data" Problem

1. The users have no assurance the private records will be handled safely

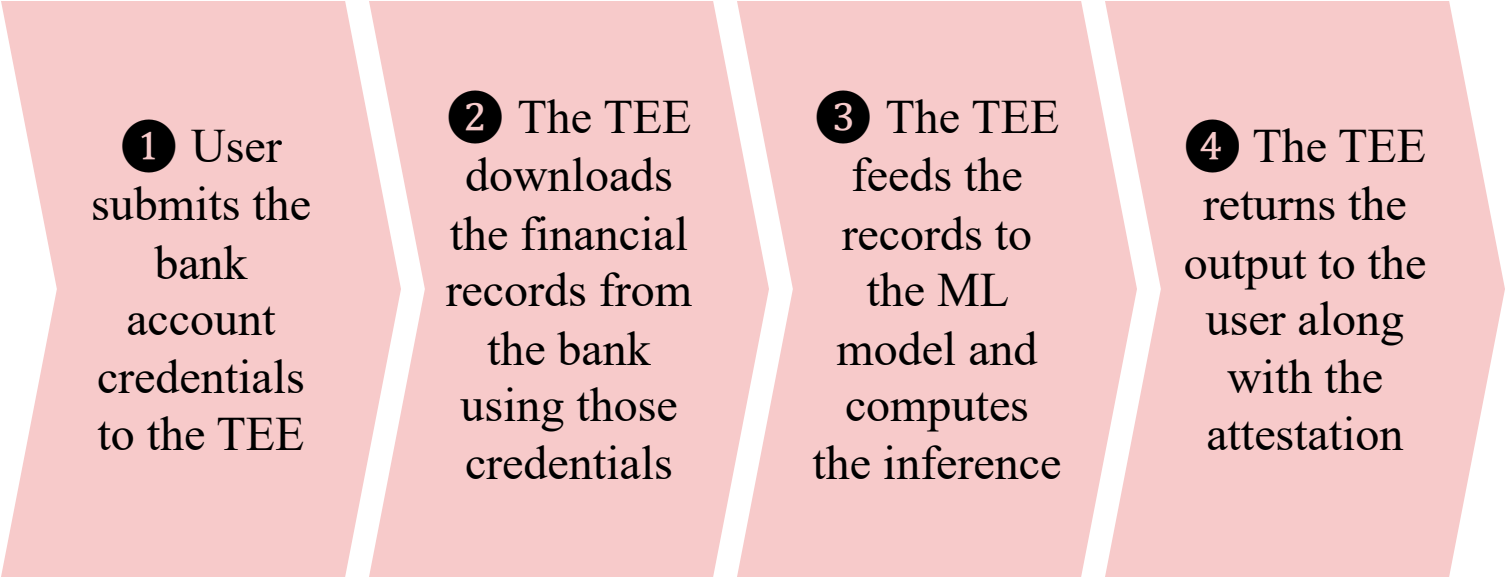
2. The services have no assurance the records were not tampered with by the user

The "Private Data" Problem

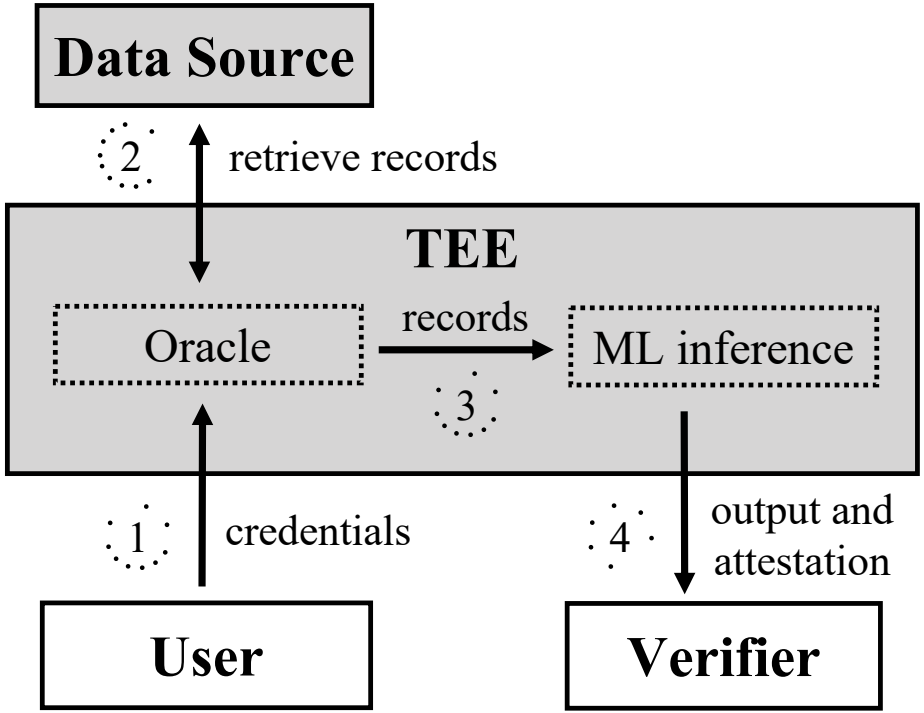
1. The users have no assurance the private records will be handled safely: **TEE privacy + attestation**

2. The services have no assurance the records were not tampered with by the user: **Oracles**

Props for Inference: Architecture



Props Architecture



Props for micro-credential

Verifiable credentials allow claims about a subject to be verified without relying on a central authority each time they are presented.

Problems with current credentials

They require a central trusted authority to generate (e.g. EUID)

OR

They can only verify simple claims due to ZKPs (e.g. CanDID)

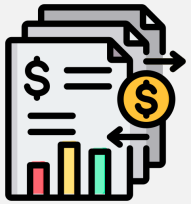
Props solves that!

1. Props does not require any architecture update from the data source.

2. LLMs and ML models allow the creation of micro-credentials over esoteric and flexible sources of data.

Some micro-credentials

EXPERTISE IN A SPECIFIC TOPIC



PROOF OF PURCHASE

HEALTH AND LIFESTYLE CLAIMS

LANGUAGE PROFICIENCY

Thank you!

https://arxiv.org/pdf/2410.20522